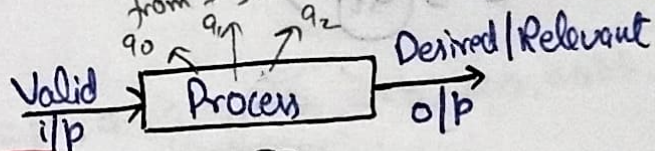
Automata Theory

- Finite Automata → Reg. Expr. / Grammar & Lang.
- Push down "
- Linear bounded "
- Turing Machine

Transition of m/c from 1 state to another



$F.A. = <Q, \Sigma, \delta, q_0, f>$

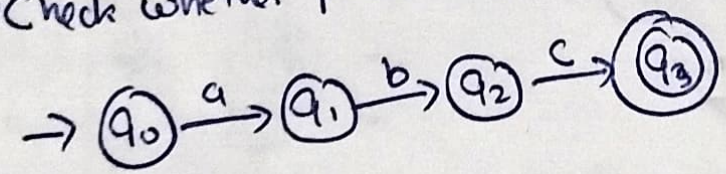
Machine → 2 states → ON / OFF

Finite Automata

↳ components of m/c are finite

→ M/c changes state provided on i/p only & not implicitly

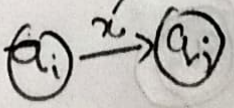
→ Check whether password (abc) is correct



→ Acceptance state { Only be reached if input string = abc else not }

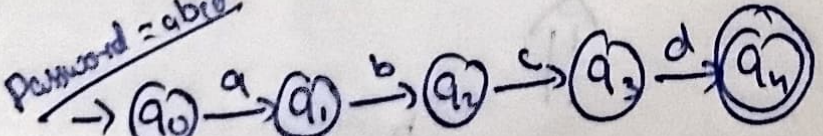
↑ Initial State

Transition Diagram Representation



valid input for any state q_i

Password = abcd



works on infinite strings but validate only 1

Terminology

Q → Finite Non-Empty set (FNES) of states

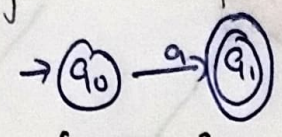
Σ → FNES of i/p symbols (alphabets)

δ → FNES of transition states $\quad \delta = \{(q_0, a) = q_1, (q_1, b) = q_2, (q_2, c) = q_3\}$

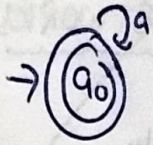
q_0 → Initial state / $q_0 \in Q$

F → FNES of final states / $f \subseteq Q$

⇒ Design a m/c for a only

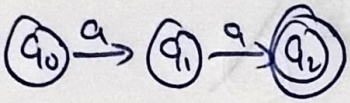


[Only a] ✓



[a as well as aa, aaa, ...] ✗

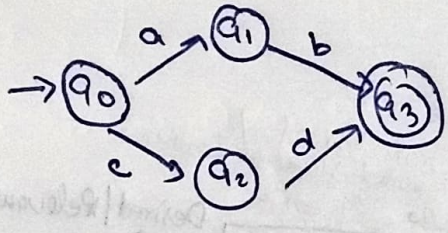
⇒ aa:



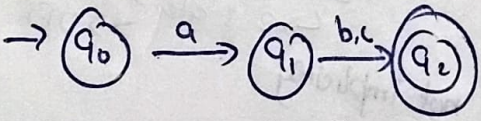
concatenation (x followed by y)

$$\begin{cases} x.y \neq y.x \\ x+y = y+x \end{cases}$$

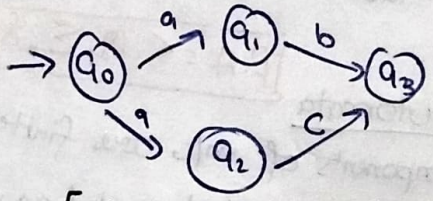
⇒ ab + cd {Either ab or cd}



⇒ ab + ac



[Deterministic] DFA

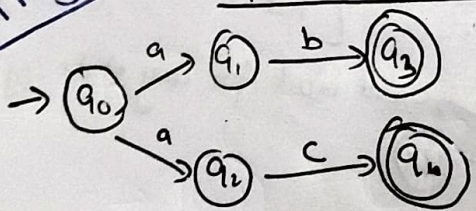


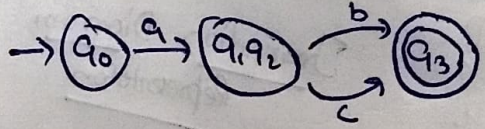
[Non-deterministic] NFA

Two transitions for same i/p

Tuesday 11 Aug, 2026

NFA to DFA conversion

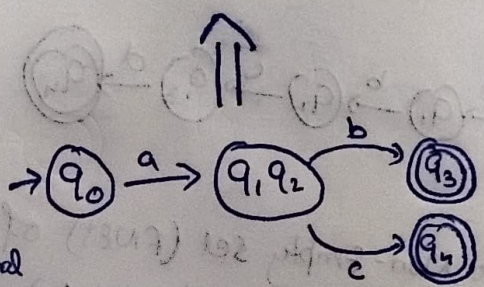




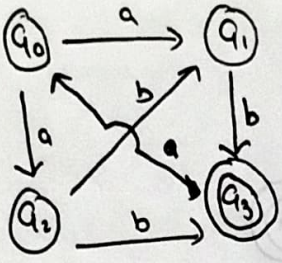
	a	b	c
q_0	q_1, q_2	-	-
q_1	-	q_3	-
q_2	-	-	q_4
q_3	-	-	-
q_4	-	-	-
q_1q_2	-	q_3	q_4

} Same rows, both final hence reduced to one state

⇒

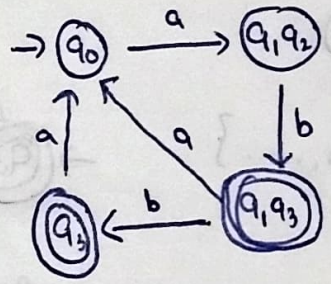


⇒

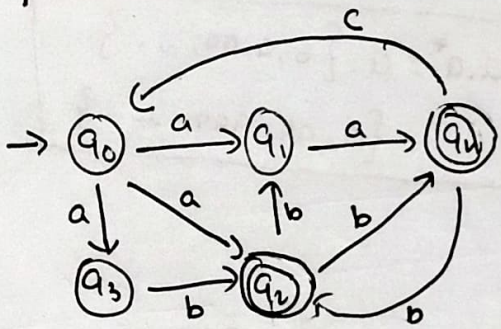


$L = \{ab, abb, abaab, abbaab \ldots\}$ that
↳ Set of strings formed by the rules of a language

	a	b
q_0	q_1, q_2	–
q_1	–	q_3
q_2	–	q_1, q_3
q_3	q_0	–
q_1, q_2	–	q_1, q_3
q_1, q_3	q_0	q_3

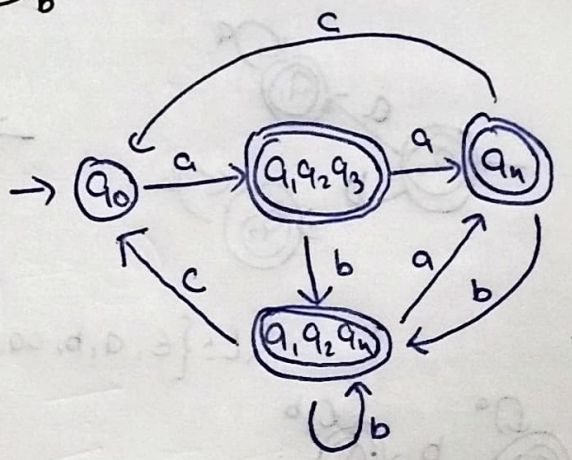


⇒



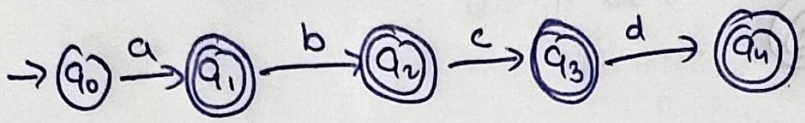
$L = \{a, aa, ab, aaca, aab, aba, aabb, abb, abca, abbca, \ldots\}$

	a	b	c
q_0	q_1, q_2, q_3	–	–
q_1	q_4	–	–
q_2	–	q_1, q_4	–
q_3	–	q_2	–
q_4	–	q_2	q_0
q_1, q_2, q_3	q_4	q_1, q_2, q_4	–
q_1, q_4	q_4	q_2	q_0
q_1, q_2, q_4	q_4	q_1, q_2, q_4	q_0



Wednesday
12 Aug, 2026

⇒ a + ab + abc + abcd



Null string : ϵ (Empty)



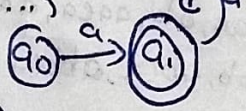
⇒ $a^* = \{\epsilon, a, aa, aaa, \ldots\}$ (Kleene's closure)



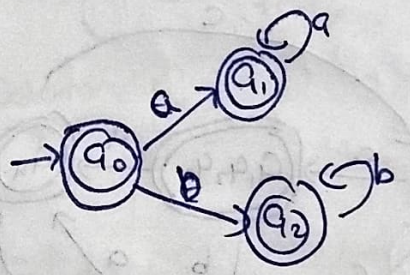
a* means 0 or more combinations of a.

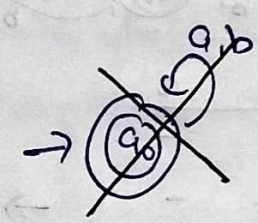
$a.a^* = a.\{\epsilon, a, aa, \ldots\}$
$= \{a, aa, aaa, \ldots\}$

⇒ $a^+ = \{a, aa, aaa, \ldots\} = aa^*$ (Positive closure of a)



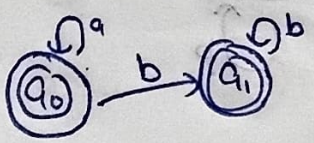
⇒ $a^* + b^*$



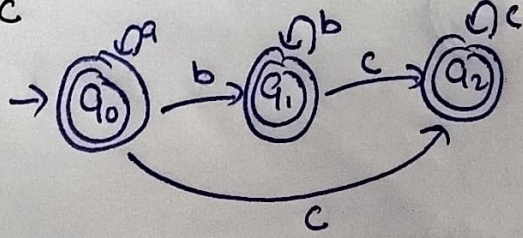


L = {ϵ, a, b, aa, bb, ab, aab, abb .... }

⇒ a^*b^*



⇒ $a^*b^*c^*$



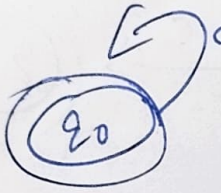
L = {ϵ, a, b, c, aa, bb, cc, ab, ac, bc, aab, abc, acc, ...}

17/8/26

Automata Theory

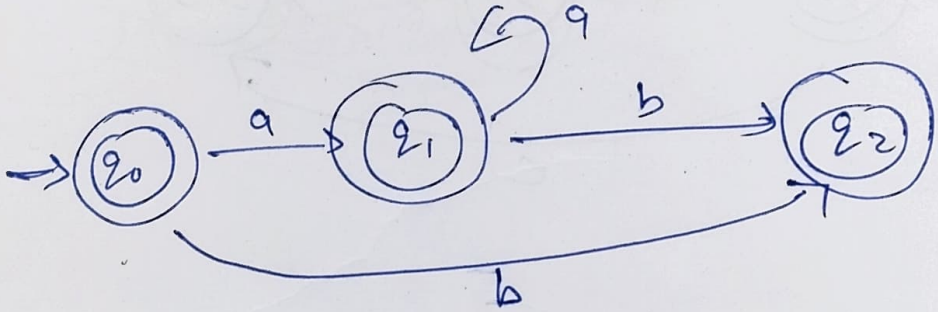
$a^* + b^*$

Ex: a^*, $a^* b^*$, [illegible], $a^* b^* c^*$, $a^* + b^*$



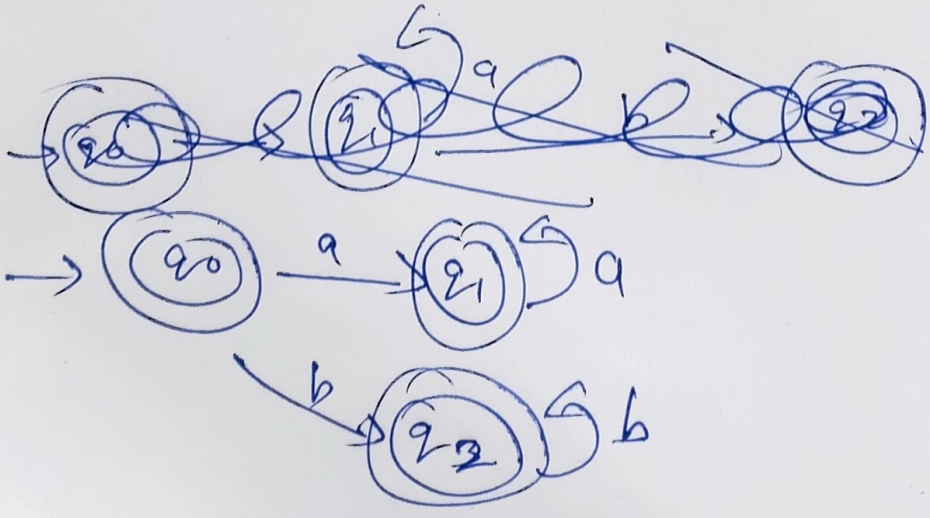
$a^* = \{\epsilon, a, aa, aaa, \ldots\}$

$a^* b^* = \{\epsilon, a, b, aa, bb, ab, aaa, bbb, abb, aab, \ldots\}$





$a^* + b^* = \{\epsilon, a, b, aa, bb, aaa, bbb \ldots\}$



$a^*(b^*+c^*)$ $L = a^* = \{\epsilon, a, aa, bb \ldots \ldots\}$

$b^* = \{\epsilon, b, bb, bb \ldots\}$

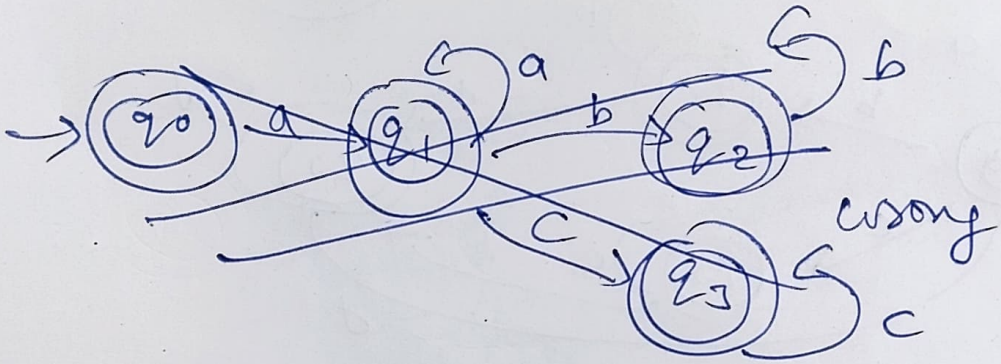
$c^* = \{\epsilon, c, cc, ccc \ldots\}$

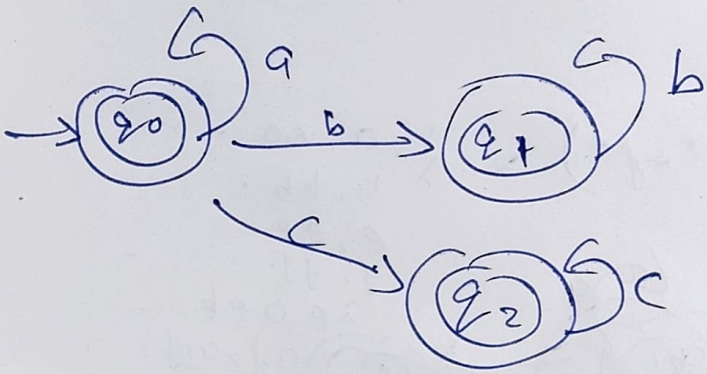
$L = \{\epsilon, a, aa \ldots \ldots$

$b, bb \ldots \ldots$

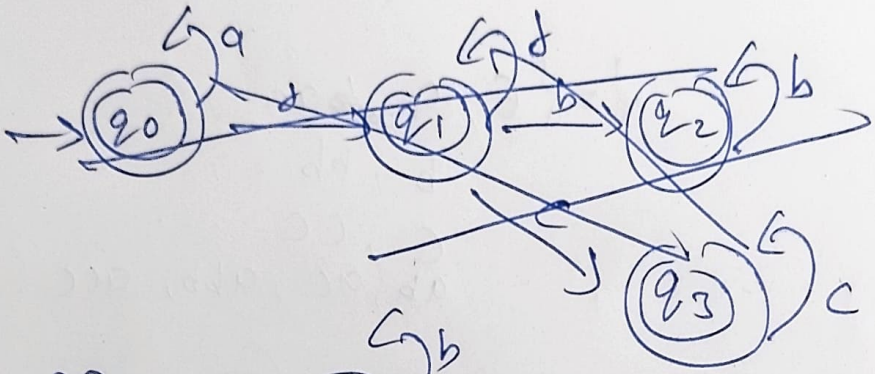
$c, cc \ldots$

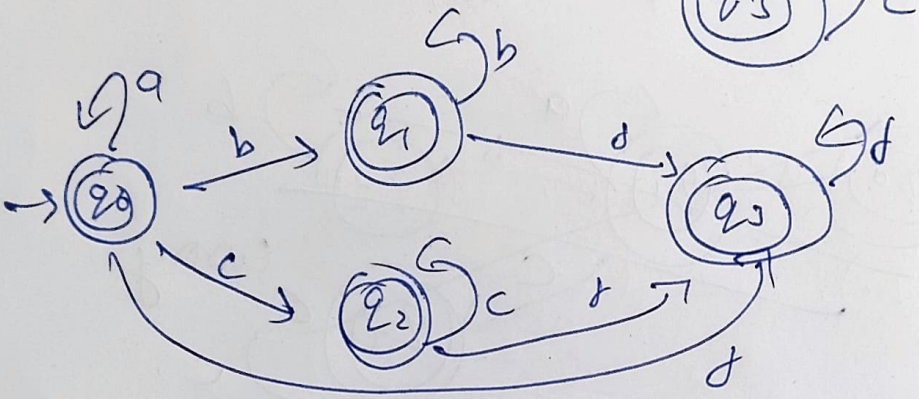
$ab, ac, abb, acc \ldots \ldots\}$



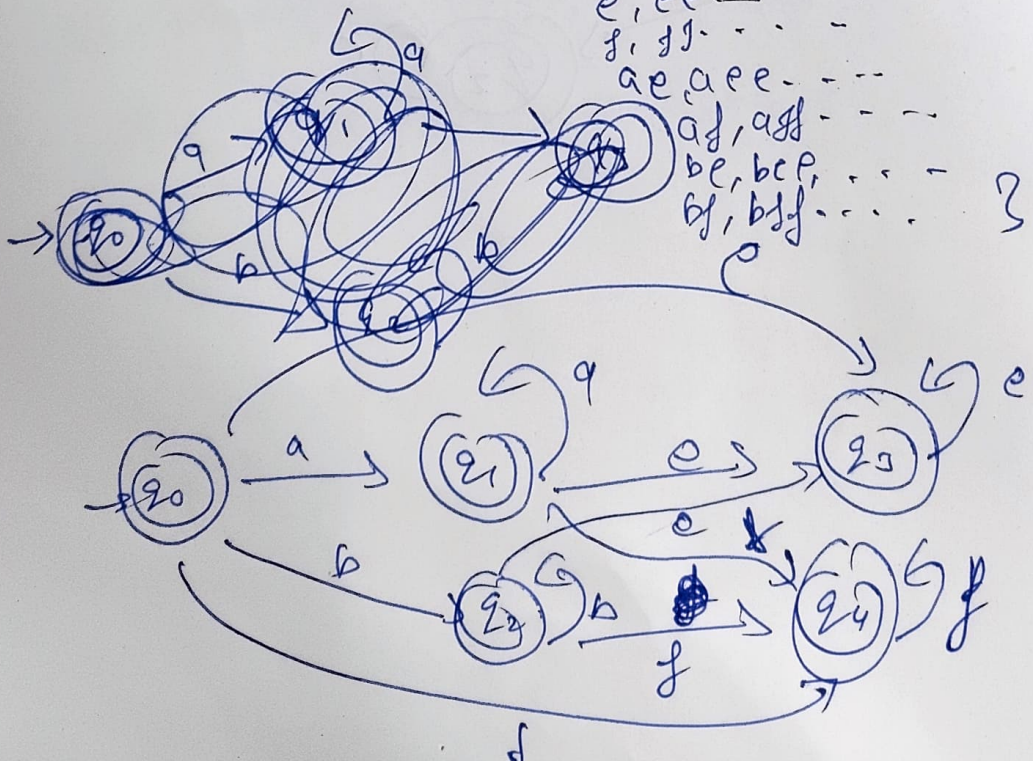


c) $a^*(b^*+c^*)d^*$ = { ε, a, aa . . .
b, bb . . .
c, cc, . . .
adb, adbb . . .
adc, adcc . . . }





d) $(a^*+b^*).(e^*+f^*)$ = { ε, a, aa . . .
b, bb . . .
e, ee . . .
f, ff . . .
ae, aee . . .
af, aff . . .
be, bee, . . .
bf, bff . . . }

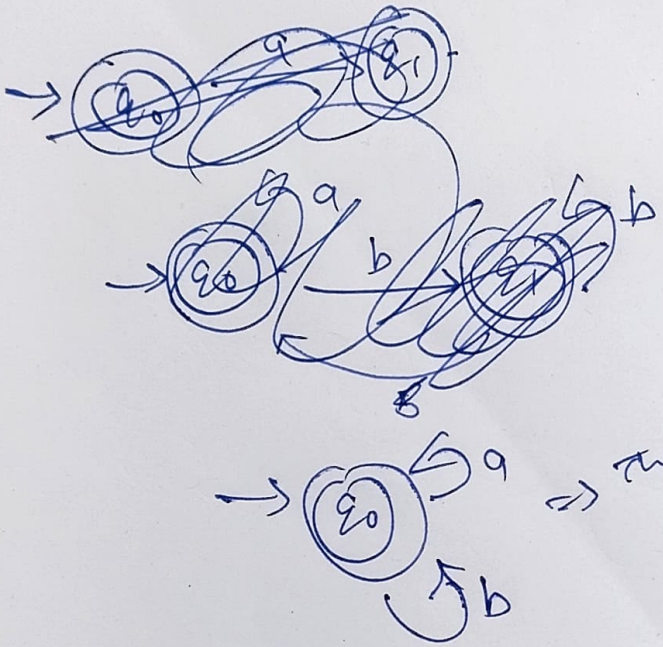


(A)

d) $(a+b)^*$ → {ε, a, aa, aaa ... b, bb, bbb ... ab, ba, aab, baa, abb, bba, ...}

⇓ any combination

$(a+b).(a+b)(a+b)(a+b)$

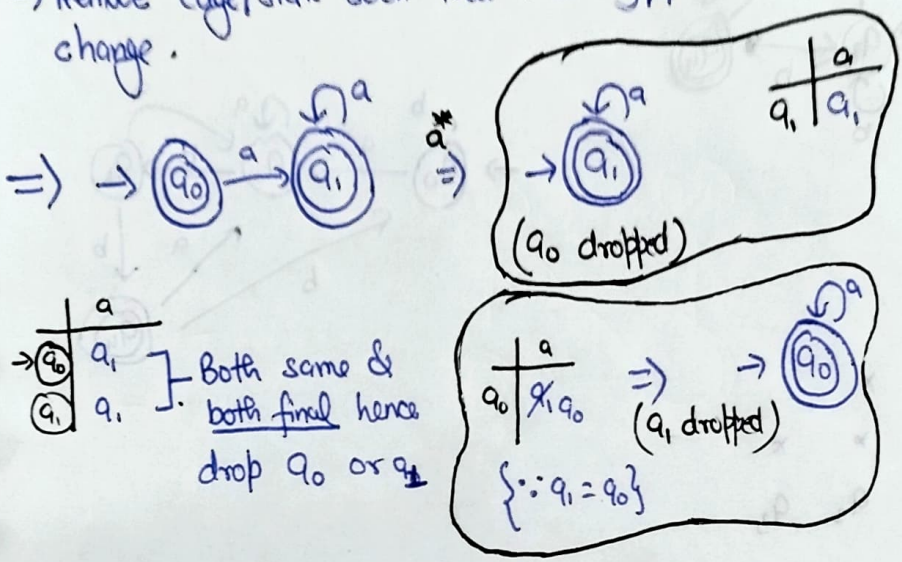


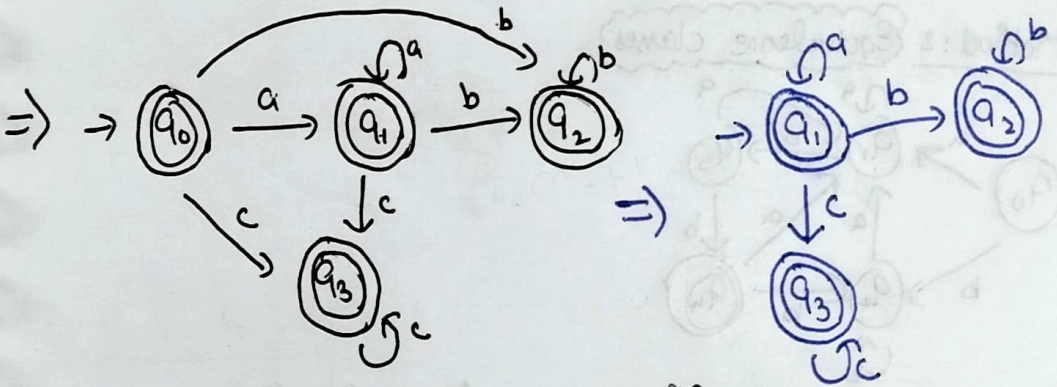
⇒ This makes any combination of $(a+b)^*$ ie $ab \neq ba$ they are different.

Tuesday
18 Aug, 2026

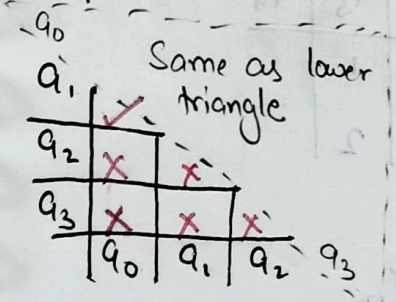
Minimization of FA

→ Remove edge/state such that meaning/functionality of FA does not change.





Comparison Matrix:



Comparison with q_0

	a	b	c
q_0	q_1	q_2	q_3
q_1	q_1	q_2	q_3
q_2	–		
q_3	–		

Both same & both final

(No need to go further)

Comparison with q_1

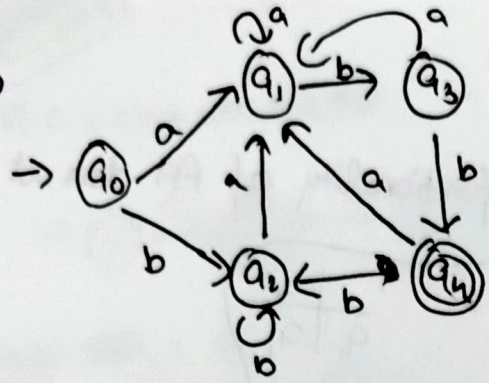
	a	b	c
q_1	q_1	q_2	q_3
q_2	–		
q_3	–		

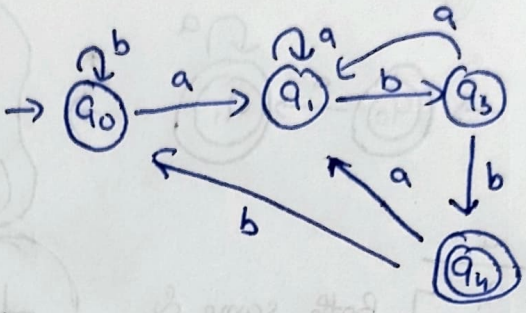
Comp. with q_2

	a	b	c
q_2	–	q_2	–
q_3	–	–	

***MATRIX METHOD OF MINIMIZATION**

⇒

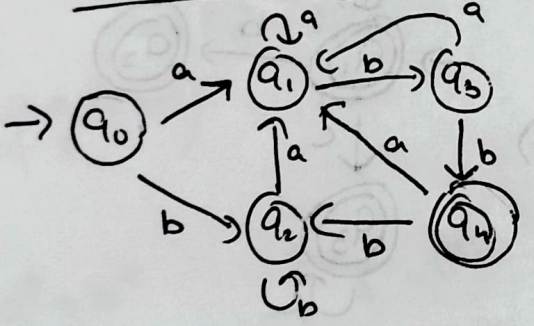




q_1	×			
q_2	✓	×		
q_3	×	×	×	
q_4	×	×	×	×
	q_0	q_1	q_2	q_3

Wednesday
19 Aug 2026

Method : 2 Equivalence classes



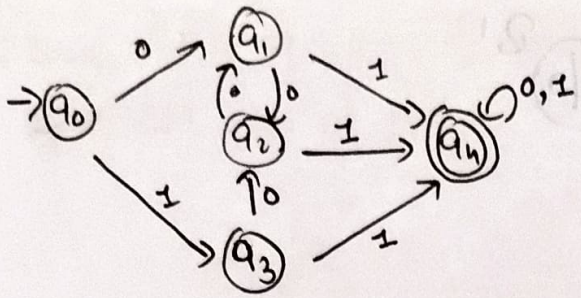
	NF ①			F ②
	q_0	q_1	q_2 q_3	q_4
a	1	1	1 1	
b	1	1	1 2	1

→

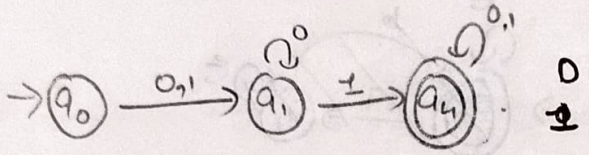
	①			②	③
	q_0	q_1	q_2	q_3	q_4
a	1	1	1	1	
b	1	2	1	2	

↓

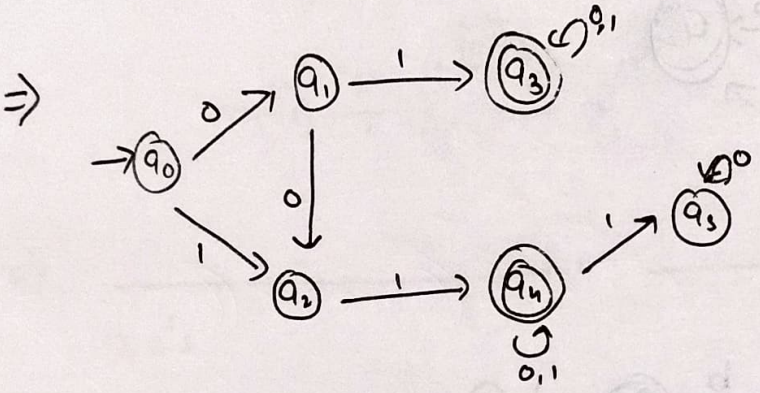
	4		①	2	3
	q_0	q_2	q_1	q_3	q_4
a	1	1			
b	4	4			



	① q_0	q_1	q_2	q_3	② q_4
0	1	1	1	1	
1	1	2	2	2	



	① q_0	② q_1	q_2	q_3	③ q_4
0		2	2	2	
1		3	3	3	



	① q_0	q_1	q_2	q_5	② q_3	q_4
0	1	1	–	1	2	2
1	1	2	2	–	2	2,1

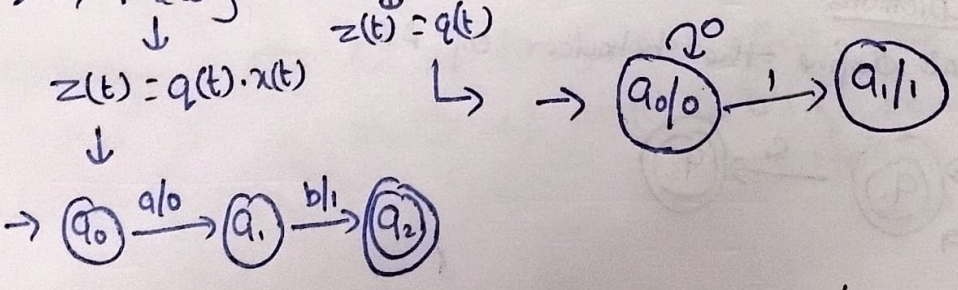
Monday
24 Aug, 2026

→ Mealy & Moore M/C } F.S.M. with o/p

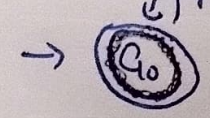
↳ Doesn't tell whether string accepted or not, rather generate string.

Mealy: $z(t) = q(t) \cdot x(t)$

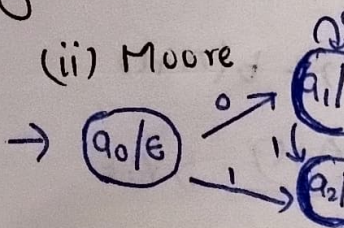
Moore: $z(t) = q(t)$



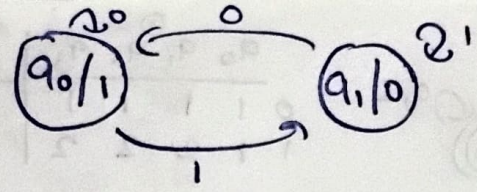
⇒ Design a Mealy M/c that generate 1's complement of given i/p.



(ii) Moore



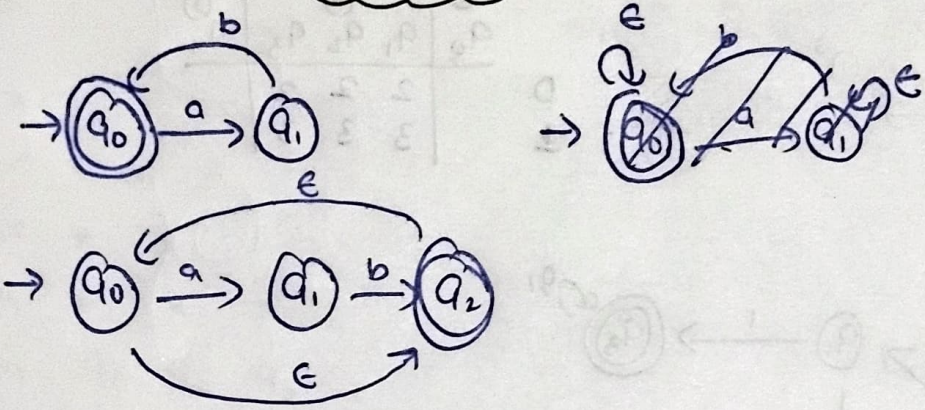
Moore:



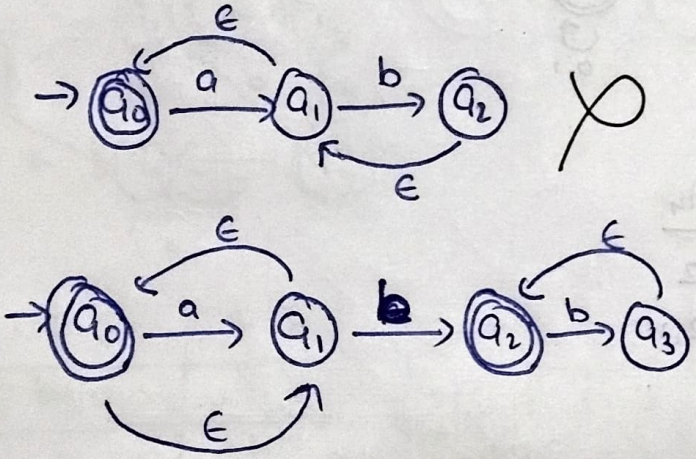
1100 → 10011

Epsilon NFA

$\Rightarrow (ab)^*$



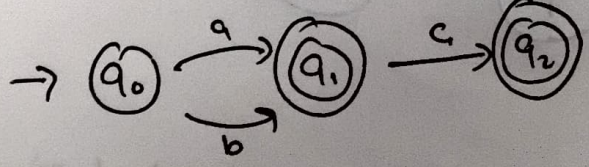
$\Rightarrow a^* b^*$



Tuesday
25 Aug, 2026

Regular Expressions

Expressions that define the behaviour of FA



$= a + ac + b + bc$

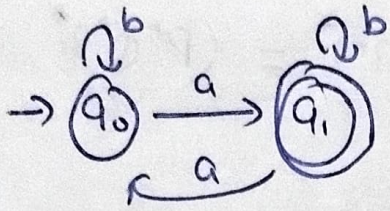
$= a(\wedge + c) + b(\wedge + c)$

$= (a+b)(\wedge + c)$ → R.E.

$\wedge \sim \epsilon$ (Epsilon)

⇒ Design a F.A. for $\Sigma\{a, b\}$ such that each string of FA has even/ odd no. of a's

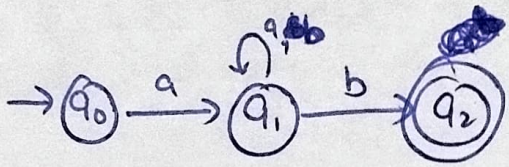
$L = \{b^*ab^*, b^*aaab^*, \ldots\}$
$b^*ab^*ab^*ab^*, \ldots$



⇒ ——— " ——— such that each string of FA starts with a & ends with b.

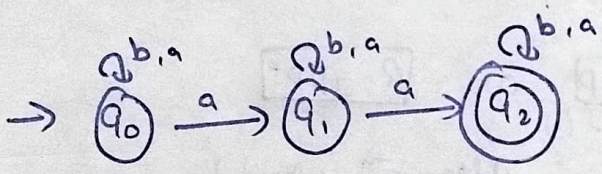
$L = \{ab, aab, aabb, abab, \ldots\}$

$a(a+b)^*b$



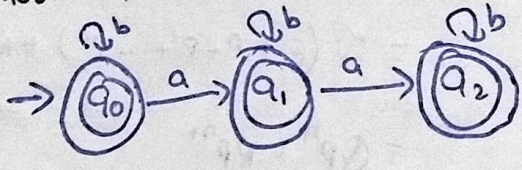
⇒ ——— " ——— has atleast 2 a's

$L = \{aa, aba, baa, aab, baba, babab, \ldots\}$



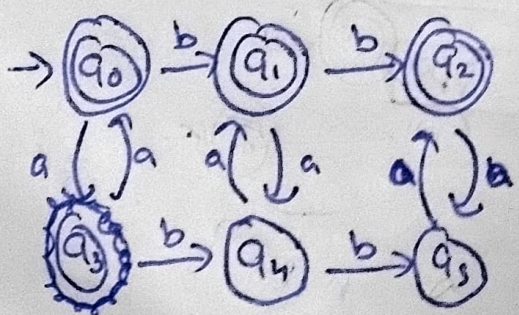
⇒ Atmost 2a's

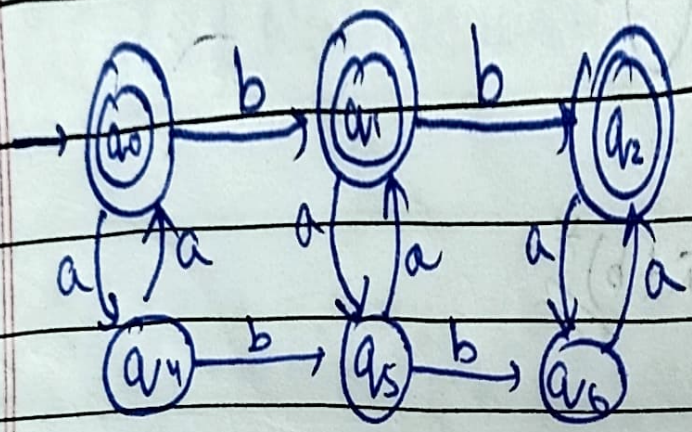
$L = \{\epsilon, b, a, ab, aab, bab, \ldots\}$



⇒ Design a DFA for $\Sigma = \{a, b\}$ such that each string contains even no. of a's & at most 2 b's. abba, abab

$L = \{\epsilon, b, bb, aabb, baab, bbaa, baaaab, \ldots\}$

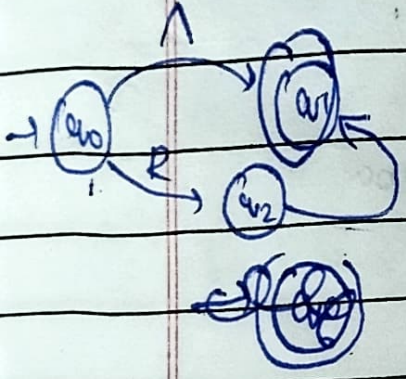


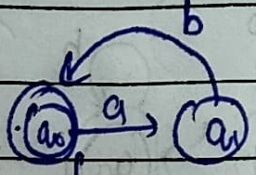


$(b^*(ab^*a)b^*)^*$

Regular exp^n → $b^*(ab^*a)^*b^*$.

→ $(b^*a\ ab^*)^*$





q_0 on b goes to ϕ not —.

Identities:-

1) $\wedge + R = R + \wedge \neq (\varepsilon / \wedge) / R$.
2) $\wedge . R = R . \wedge = R$
3) $\phi + R = R + \phi = R$
4) $\phi . R = R . \phi = \phi$.
5) $(R^+)^* = R^*$
6) $RR^* = R^*R = R^+$.
7) $\wedge + RR^* = R^*$
8) $QP + RP = P(Q+R)$ P

DFA.txt.

0	→ Initial
0,1	→ final
1 1	
-1 0	

ϕ → No

q_0 12 $\wedge$ → no string

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- $(Q+P)^* = (Q^* + P^*)^* = (Q^*P^*)^* = (P^*Q^*)^*$

  $\Sigma_1 = \{\epsilon, Q, P, QQ, PP, QPQ, PQP, \ldots\}$

  $\Sigma_2 = \Sigma_1$

  $\Sigma_3 = \{\epsilon, Q, P, QP, PQ, QQ, PP, QPQ, PQP, \ldots\}$

  $\Sigma_4 = \Sigma_3$

- $P(QP)^* = (PQ)^*P$

  $\Sigma_1 = \{P, PQP, PQPQP \ldots\}$

  $\Sigma_2 = \{P, PQP, PQPQP \ldots\}$

Arden's Theorem

We can rewrite $\boxed{R = Q + RP}$ - ① as $\boxed{R = QP^*}$

$R = Q + RP$

$= Q + (Q + RP)P$

$= Q + QP + RP^2$ - ②

$= Q + QP + (Q + RP)P^2$

$= Q + QP + QP^2 + RP^3$ - ③

After n^{th} substitution

$R = Q + QP + QP^2 + \ldots QP^n + RP^{n+1}$

$= Q(\epsilon + P + P^2 + \ldots P^n) + RP^{n+1}$

$= QP^n + RP^{n+1}$ $\{\forall\ n \geq 0\}$ $\{P^n = P^*\}$

↳ {useless as $RP^{n+1} \subset QP^n$}

$\Rightarrow \boxed{R = QP^n}$

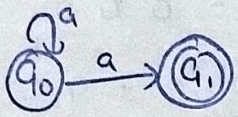
$\boxed{R = QP^*}$

⇒ Steps to convert F.A. to Reg. Exp. :

1.) Construct eq. of each state of F.A.

2.) Use these eq. to evaluate the final state-free expression of the final state using substitution method.

3.) The eq. of the final state is the eq. of the automaton.

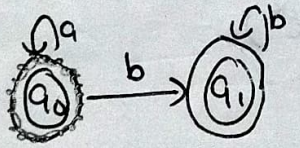
⇒ 

1.) $q_0 = q_0a + \wedge = \wedge(a)^* = a^*$

$q_1 = q_0a$

2.) $q_1 = q_0a = a^*a$

3.) $\boxed{q_1 = a^*a}$

⇒ 

1.) $q_0 = q_0a + \wedge = \wedge a^* = a^*$ -(i) {Using Arden's theorem}

$q_1 = q_0b + q_1b = q_0b(b)^* =$ ~~q_0b^*~~ -(ii)

2.) $q_1 =$ ~~a^*b^*~~ {Substituting eqn (i) in eqn (ii)}

$= (a^*b)b^*$

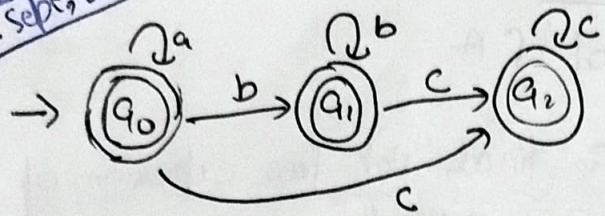
If q_0 is also final:

$q_f = q_1 + q_0 = a^*bb^* + a^*$

$= a^*(bb^* + \wedge)$

$= a^*b^*$ $\{\because \wedge + RR^* = R^*\}$

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$q_0 = q_0a + \wedge \quad = \wedge a^* = a^*$

$q_1 = q_0b + q_1b \quad = a^*b + q_1b = a^*b.b^*$

$q_2 = q_0c + q_1c + q_2c = a^*c + a^*bb^*c + q_2c$

$= (a^*c + a^*bb^*c).c^*$

$= a^*(\wedge + bb^*)cc^* \quad \{\because \wedge + RR^* = R^*\}$

$= a^*b^*cc^*$

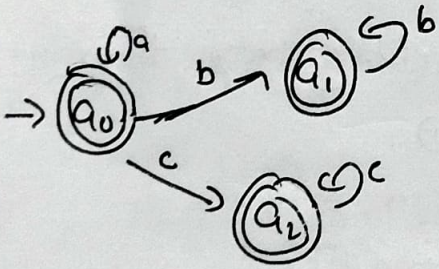
Using arden's theorem

$q_f = a^* + a^*bb^* + a^*b^*cc^*$

$= a^*(\wedge + bb^* + b^*cc^*) \quad \{\because \wedge + RR^* = R^*\}$

$= a^*(b^* + b^*cc^*) = a^*b^*(\wedge + cc^*)$

$= a^*b^*c^*$



$q_0 = q_0a + \wedge = \wedge a^* = a^*$

$q_1 = q_0b + q_1b = a^*b + q_1b = a^*bb^*$

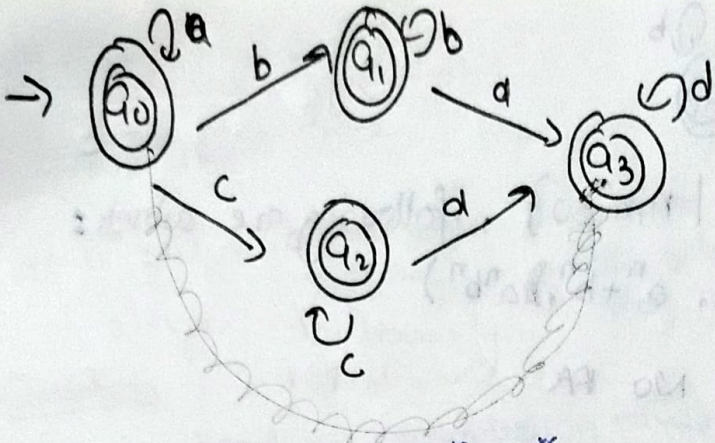
$q_2 = q_0c + q_2c = a^*c + q_2c = a^*cc^*$

Using arden's theorem

$q_f = q_0 + q_1 + q_2 = a^* + a^*bb^* + a^*cc^*$

$= a^*(\wedge + bb^* + cc^*) \quad \{\because \wedge + RR^* = R^*\}$

$= a^*(b^* + c^*)$



$q_0 = q_0a + \wedge = \wedge a^* = a^*$

$q_1 = q_0b + q_1b = a^*b + q_1b = a^*bb^*$

$q_2 = q_0c + q_2c = a^*c + q_2c = a^*cc^*$

$q_3 = q_1d + q_3d + q_2d = a^*bb^*d + q_3d + a^*cc^*d$

$= (a^*bb^*d + a^*cc^*d)d^*$

$= a^*(bb^* + cc^*)dd^*$

$q_f = q_0 + q_1 + q_2 + q_3 = a^* + a^*bb^* + a^*cc^* + a^*(bb^* + cc^*)dd^*$

$= a^*(\wedge + bb^* + cc^*) + a^*(bb^* + cc^*)dd^*$

{Not optimal step} ← $= a^*(b^* + c^*) + a^*(bb^* + cc^*)dd^*$

$= a^*(b^* + bb^*dd^* + c^* + cc^*dd^*)$

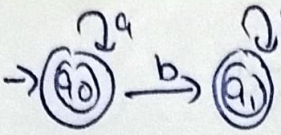
$= a^*(\wedge + bb^* + bb^*dd^* + \wedge + cc^* + cc^*dd^*)$

$= a^*(\wedge^+ bb^*(\wedge + dd^*) + \wedge^+ cc^*(\wedge + dd^*))$

$= a^*(\wedge^+ bb^*d^* + \wedge^+ cc^*d^*)$

$= a^*(\wedge + bb^* + \wedge + cc^*)d^*$

$= a^*(b^* + c^*)d^*$

$a+b^* \Rightarrow$ 

for a language, $L = \{a^n b^m \mid n, m \geq 0\}$, following are subsets:

$(a^n, b^n, a^n + b^n, a^n b^n)$

↳ context-free language
↳ requires PDA.

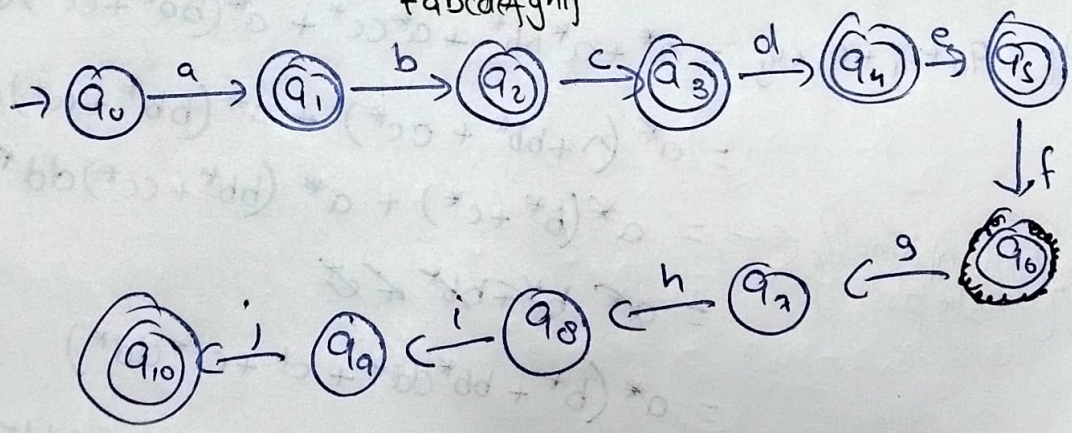
$a^n b^n \rightarrow$ No FA

- Finite automaton works on regular languages (that contain repetition) but $a^n b^n$ requires recursion, hence limits F.A.

- 4 language types: [Chomsky classification]
  (i) Regular
  (ii) Context-free
  (iii) Context-sensitive
  (iv) Unbounded

Monday 7 Sept, 2026

$\Rightarrow L = \{a + ab + abc + abcd + abcde + abcdefghij\}$



Max states = longest length sequence + 1

Pumping Lemma is used to prove non-regularity of languages.

- Let there be k-states
- for ex- $L = \{a^n b^n \mid n \geq 0\}$
  $\Rightarrow w = a^k b^k \Rightarrow$ length $= 2k$
- Divide w into xyz such that x, z may or may not be ϵ & y should repeat {0 to ∞}

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9 Sept, 2026

$\Rightarrow \{a^n, n \geq 0\}$

Let $a^k = \omega$ = aaa.....a^k $\Rightarrow$ y = 0 to ∞
x y z
all belongs to a^n
Hence, test passed

Conditions

- P.L. is based on contradiction theory where the language under question is assumed to be regular & having a finite automaton with k no. of states.
- Now we select a word 'ω' such that $\omega \in L$ & $|\omega| \geq k$
- We decompose ω into 3 components – x, y & z such that the following 3 conditions hold:
  (i) $|xy| \leq k$
  (ii) $y \neq \epsilon$ or $\{|y| = p \mid p > 0\}$
  (iii) $xy^iz \in L$ ∀ $i \geq 0$
- If these 3 conditions hold ∀ values of i, the lang. is regular.

$\Rightarrow \{a^n b^n \mid n \geq 0\}$

Let $\omega = a^k b^k = xyz$

Since $|xy| \leq k \Rightarrow xy = aa....a^k$

$\rightarrow$ Let $y = a^p \Rightarrow x = a^{k-p}$
$z = b^k$

Now, xy^iz must belong to L
but for i=0, $xy^iz = xz = a^{k-p}b^k \notin L$ → Hence non-regular

$\Rightarrow a^*b^*$

Let $\omega = a^k b^k = xyz$

$|xy| \leq k \Rightarrow xy = aa...a^k$

Let $y = a^p \Rightarrow x = a^{k-p} \Rightarrow z = b^k$

Now, $xy^iz \in L$
If i=0 $\Rightarrow xz = a^{k-p}b^k \in L$
i=1 $\Rightarrow xyz = a^kb^k \in L$
i=2 $\Rightarrow xy^2z = a^{k+p}b^k \in L$
⋮
Hence, we can say that this lang. may be regular.

$\Rightarrow \{a^n b^m c^n \mid n, m \geq 0\}$

$\omega = a^k b^k c^k \Rightarrow |\omega| = 3k$

$|xy| \leq k = aaa\ldots a^k$

$\Rightarrow$ If $|y| = p \Rightarrow x = a^{k-p}, y = a^p, z = b^k c^k$

$\omega' = xy^i z$

$i = 0 \Rightarrow xz = a^{k-p} b^k c^k$

Since $|a| \neq |c|$

Hence $xy^i z \in L$ at $i = 0$

Hence, non-regular.

$\Rightarrow \{a^n \mid n \geq 0$ & n is prime$\}$

Let $\omega = a^5 \Rightarrow |\omega| = 5$

$|xy| \leq k = 5$

Let $y = a \Rightarrow x = a^4 \Rightarrow z = \epsilon$

$\Rightarrow \omega' = xy^i z$

at $i = 0$

$\omega' = xz = a^4 \notin L$

Hence, given language is non-regular

Sessional - 1 Syllabus Complete